IIAD Users Manual

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1. Introduction

In recent years, the detection of ionospheric irregularities using the total electron content (TEC) change rate index (ROTI) of GNSS has become a popular research direction. However, there are few open-source programs in this field. Here, we have developed a software IIAD based on Python and MATLAB for identifying ionospheric irregularities. IIAD is capable of handling multi-system GNSS, including GPS, GLONASS, Galileo and BDS and form different dual-frequency combinations to calculate TEC. Based on the calculated TEC, ROTI can be calculated. The extracted ROTI value, after being trained by machine learning, can identify ionospheric irregularities.

2. Run

The main process of ionospheric irregularity using IIAD software is shown in Figure 1.

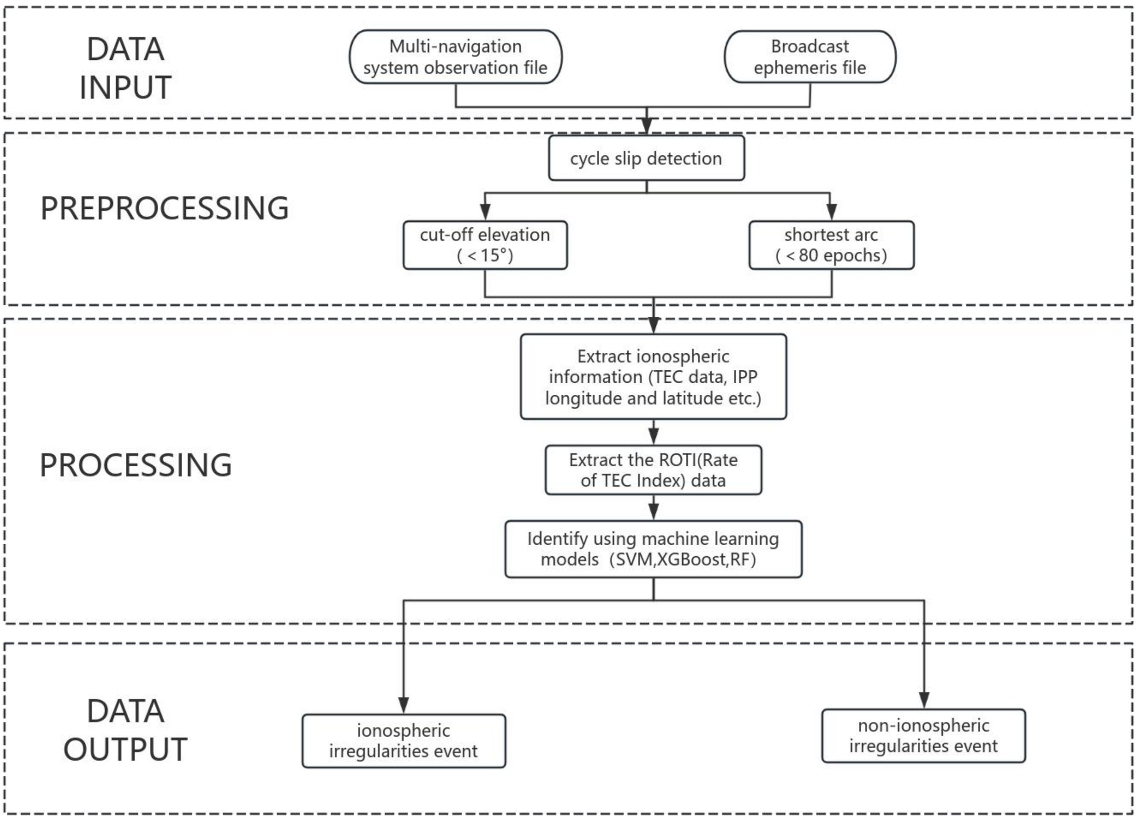

Fig. 1 Operation flowchart of IIAD

2.1 Generate TEC files

This is accomplished by the *getTEC* function. The main options are list in Table 1. The processing options include the type of dual-frequency geometry-free combination, the navigation system, the elevation mask, the thresholds of cycle-slip detection and the height of ionosphere shell. The input files include GNSS observation and navigation files. The output files are TEC files with filename extension '. tec' for each GNSS station. Table 2 shows the instruction for the TEC file.

Table 1 The options for the script *getTEC*.

Struct opt	.gftyp	Type of dual-frequency geometry-free combination. The value is 1 or 2. Specifically, GPS: =1, L1&L2; =2, L1&L5 GLO: =1,G1&G2; =2, G1&G3 GAL: =1,E1&E5b;=2, E1&E5a BDS: =1,B1&B2; =2, B1&B3
	.elmask	Satellite elevation mask in degree.
	.navsys	Type of navigation system (char). Specifically, 'G' represents GPS, 'R' represents GLONASS, 'E' represents Galileo and 'C' represents BDS. Any combination of navigation system is feasible. For example, the value 'GE' indicates GPS and Galileo are selected.
	.gfslipthres	Threshold of cycle slip detection by geometry-free combination in meter
	.mwslipthres	Threshold of cycle slip detection by Melbourne-Wubbena combination in meter.
	.hion	Height of ionospheric shell in meter.
I\O	obsfile	Input GNSS observation files path and name. The wildcard is supported.
	navfile	Input GNSS navigation files path and name. The wildcard is supported.
	outpath	Output TEC file folder

Table 2 Instruction of the output TEC file

Column	Instructions
DAY	Date (with format 'yyyymmdd')
SEC	Seconds in a day
NSAT	Satellite number (integer)
PRN	Satellite PRN (char)

NSTA	Station number (integer)
NAME	Station name
STEC	Slant TEC
MF	Mapping factor (<=1)
PLAT	Latitude of ionospheric piece point in degree
PLON	Longitude of ionospheric piece point in degree

2.2 Generate ROTI files

This is accomplished by the *getROTI* function. Table 3 lists the main options. The input file is a TEC file, and the file extension is '.tec'. The output file is a ROTI file with the file extension '.rot'. Table 2 shows the instructions of the ROTI file.

Table 3 The options for the interface of getROTI.

Struct opt	.et	Sampling rate in minute
	.wt	The sliding time window size in minute of the calculation of ROTI
I\O	tecfile	Input GNSS TEC files path and name. The wildcard is supported.
	outpath	Output ROTI file folder

Table 4 Instruction of the output ROTI file

Column	Instructions
DAY	Date (with format 'yyyymmdd')
SEC	Seconds in a day
PRN	Satellite PRN (char)
NAME	Station name
ROT	Rate of TEC
ROTI	Rate of TEC index
MF	Mapping factor (<=1)
PLAT	Latitude of ionospheric piece point in degree
PLON	Longitude of ionospheric piece point in degree

2.3 Detection of ionospheric irregularities

This is accomplished by the *Detection* function.Table 5 lists the main options, and users can choose three models by themselves. For the input and output of the file, please refer to the following running example.

Table 5 The options for the interface of Detection.

Struct opt	XGBoost	eXtreme Gradient Boosting
	SVM	Support vector machines
	RF	Random Forest
I\O	Train file	Input Train files path and name.
	Data file	Input Date files path and name， to be detected
	Model file	Input the path and name of the model that has been trained
	outpath	Output model file folder

3. Usage method

To use all the functions of the software, you only need to install the required dependency libraries. Please refer to the file ‘requirements.txt’. Run *main.py* to use the interactive interface (GUI).

3.1 TEC test

Solving the ionospheric TEC using GNSS files is a key step in detecting ionospheric disturbances. In the software, several GNSS stations are provided for test. Table 6 lists the input files and output results.

Table 6 Input and output files of the TEC test.

Input files	Output files
ALMA3250.21o, AMTE3250.21o brdc3250.21n	ALMA3250.tec, AMTE3250.tec

When calculating TEC using GNSS observation data, you need to click the ‘getTEC’ button to open the interface shown in Fig.2, enter the folder path where the GNSS observation data file (‘o’ file) is located and the path where the GNSS navigation data file (‘n’ file) is located. Then input the Output path. The default values of ‘gftype’, ‘elmask’, ‘navsys’, ‘gfslipthres’, ‘mwslipthres’, and ‘hion’ are 1, 15, 'G', 0.5, 5, 350000, respectively. For the specific meanings of these variables, please refer to Table 1. The example of the solved TEC file is shown in Fig.3.

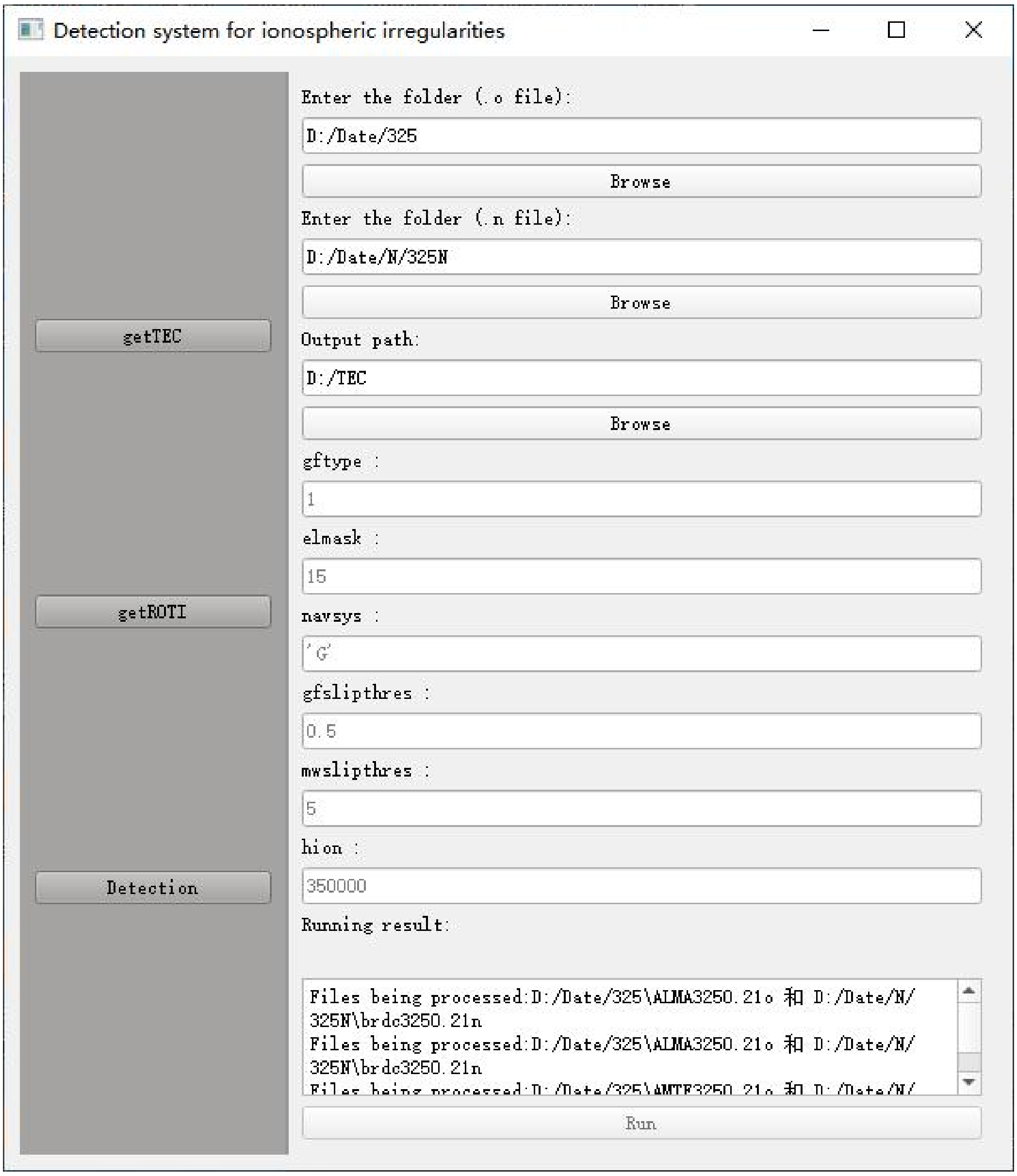

Fig. 2 The usage method of calculate TEC

```
% Program:output the STEC in IPP (including satellite/receiver DCBs) unit/TECu
%      DAY      SEC  NSAT  PRN  NSTA  NAME     STEC     MF     PLAT     PLON
  20211121     0.0    1   G01    1  ALMA    95.398   0.525  -15.097  -39.337
  20211121    30.0    1   G01    1  ALMA    95.242   0.527  -15.053  -39.310
  20211121    60.0    1   G01    1  ALMA    95.180   0.529  -15.009  -39.284
  20211121    90.0    1   G01    1  ALMA    95.043   0.531  -14.965  -39.259
  20211121   120.0    1   G01    1  ALMA    94.950   0.533  -14.922  -39.233
  20211121   150.0    1   G01    1  ALMA    94.772   0.535  -14.879  -39.208
  20211121   180.0    1   G01    1  ALMA    94.645   0.537  -14.836  -39.183
  20211121   210.0    1   G01    1  ALMA    94.464   0.539  -14.794  -39.159
  20211121   240.0    1   G01    1  ALMA    94.374   0.541  -14.752  -39.135
  20211121   270.0    1   G01    1  ALMA    94.255   0.543  -14.711  -39.111
  20211121   300.0    1   G01    1  ALMA    94.127   0.545  -14.670  -39.088
  20211121   330.0    1   G01    1  ALMA    94.010   0.548  -14.629  -39.065
  20211121   360.0    1   G01    1  ALMA    93.847   0.550  -14.589  -39.042
  20211121   390.0    1   G01    1  ALMA    93.779   0.552  -14.549  -39.020
```

Fig. 3 The generated TEC file for station ALMA on November 21, 2021.

3.2 ROTI test

ROTI is a widely used parameter for detecting the spatial scale of ionospheric disturbances. In the software, we conducted tests based on ‘ALMA3250.tec’. Table 7 lists the input files and output results.

Table 7 Input and output files of the ROTI test

Input files	Output files
ALMA3250.tec	ALMA3250.rot, ALMA3250.xlsx

When calculating ROTI using the TEC file, you need to click the ‘getROTI’ button to open the interface shown in Fig.4, enter the path where the TEC file is located, and then enter the output path. The default values of ‘rot-et’ and ‘rot-wt’ are 0.5 and 5 respectively, with the unit being min. The specific meanings of these variables are shown in Table 3. The example of the solved ROTI file is shown in Fig.5.

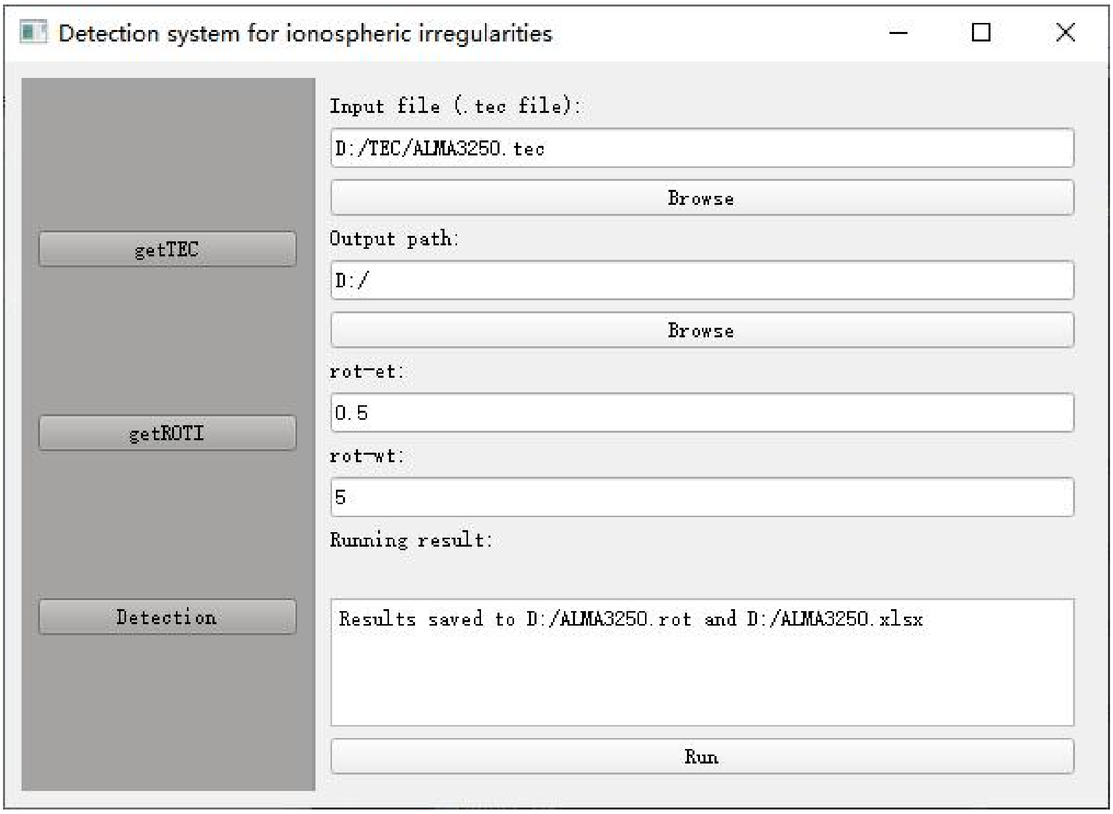

Fig. 4 The usage method of calculate ROTI

20211121	300.00	G01	ALMA	-0.256000	0.072864	0.545	-14.670000	-39.088000
20211121	330.00	G01	ALMA	-0.234000	0.070392	0.548	-14.629000	-39.065000
20211121	360.00	G01	ALMA	-0.326000	0.060683	0.55	-14.589000	-39.042000
20211121	390.00	G01	ALMA	-0.136000	0.072057	0.552	-14.549000	-39.020000
20211121	420.00	G01	ALMA	-0.306000	0.069892	0.554	-14.509000	-38.998000
20211121	450.00	G01	ALMA	-0.234000	0.063240	0.556	-14.470000	-38.976000
20211121	480.00	G01	ALMA	-0.182000	0.066676	0.558	-14.431000	-38.954000
20211121	510.00	G01	ALMA	-0.260000	0.054803	0.56	-14.392000	-38.933000
20211121	540.00	G01	ALMA	-0.096000	0.067570	0.562	-14.354000	-38.912000
20211121	570.00	G01	ALMA	-0.244000	0.067694	0.564	-14.315000	-38.891000
20211121	600.00	G01	ALMA	-0.296000	0.070393	0.567	-14.278000	-38.871000
20211121	630.00	G01	ALMA	-0.068000	0.085726	0.569	-14.240000	-38.851000
20211121	660.00	G01	ALMA	-0.094000	0.083866	0.571	-14.203000	-38.831000
20211121	690.00	G01	ALMA	-0.154000	0.082840	0.573	-14.166000	-38.811000

Fig. 5 The result of calculate ROTI

3.3 Detection test

Our software provides three machine learning models for the detection of ionospheric irregularities, namely XGBoost, SVM and RF. Users can make their own choices in the drop-down box, as shown in Fig.6. Table 8 lists the input files and output results.

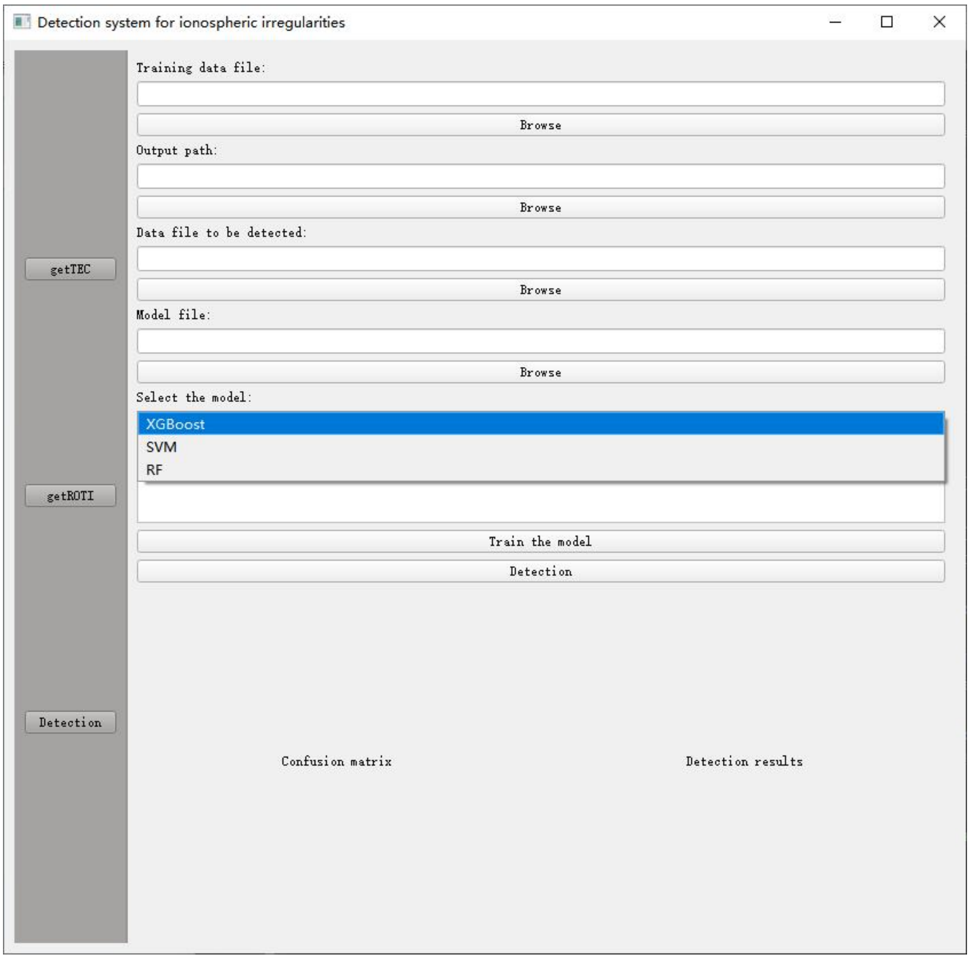

Fig. 6 Three training models

Table 8 Input and output files of the Detection test

	Input files	Output files
Training and validation	Train.xlsx	xgboost_confusion_matrix.tif svm_confusion_matrix.tif rf_confusion_matrix.tif xgboost_model.pkl svm_model.pkl rf_model.pkl
Test	xgboost_model.pkl svm_model.pkl rf_model.pkl Test.xlsx	xgboost_prediction_results.txt svm_prediction_results.txt rf_prediction_results.txt xgboost_prediction_plot.png svm_prediction_plot.png rf_prediction_plot.png

Taking the XGBoost model as an example, we tested the ionospheric irregularities detection function of IIAD. During the process of preparing the training data, the ROTI data were manually classified and classification labels were assigned. The TEC time series does not use fixed thresholds but visual inspection to determine the occurrence of ionospheric irregularities and record their start and end times. Due to its high precision, this visual inspection can directly identify all ionospheric irregular body events in the TEC time series and is helpful for manually assigning labels to ROTI. We provide users with the dataset calibrated by the visual inspection method, refer to ‘Train.xlsx’. If users want to use their own dataset, please refer to ‘Train.xlsx’ for classification. When training the model, you need to select the location of the training set and the location to save the model. Clicking the "Train the model" button will save the model and generate a confusion matrix for validation, which will be displayed on the interface as shown in Fig. 7. When detecting whether there are ionospheric irregularities in a new dataset, you need to select the data to be detected and the trained model (‘.pkl’). It is worth noting that when choosing different model files, the drop-down box needs to be kept corresponding, because the data input format of XGBoost is different from that of SVM and RF. After inputting, clicking the ‘Detection’ button will save the detection results, and the detection result image will be displayed on the interface as shown in Fig. 8.

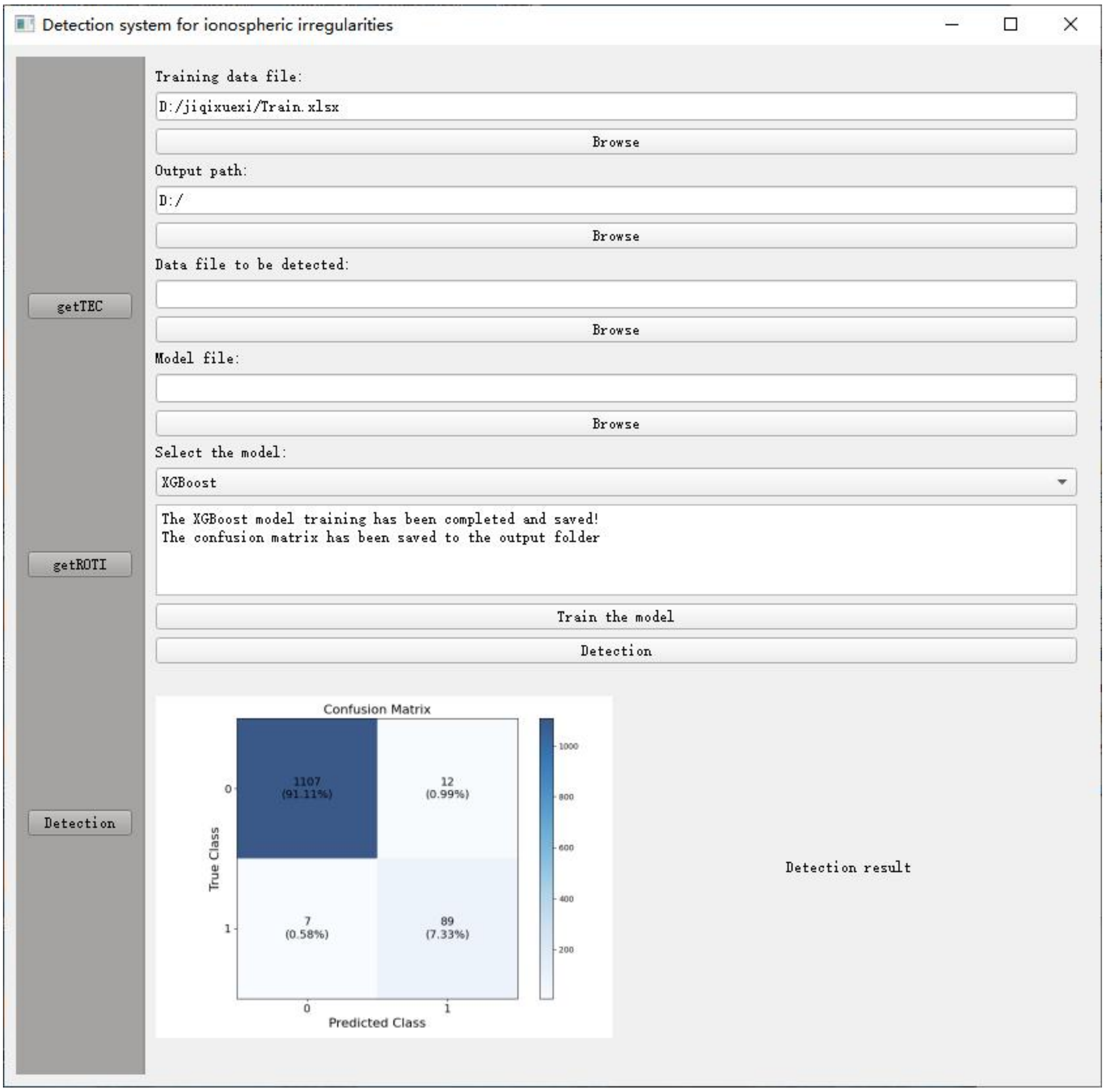

Fig. 7 Train the model.

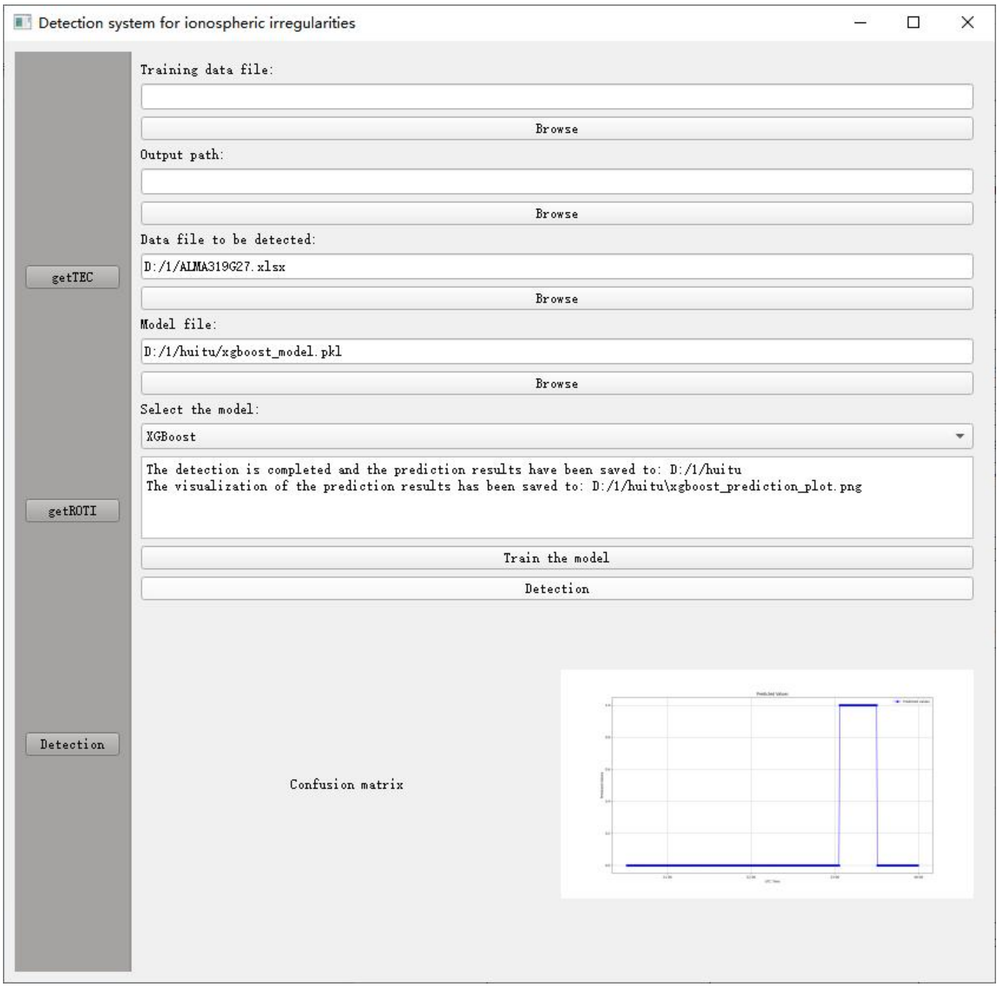

Fig. 8 Test in new dataset.

4. Methodology

The slant TEC (sTEC) is calculated as shown in Equation (1) :

$$sTEC = \left(\frac{L_1}{f_1} - \frac{L_2}{f_2}\right) \cdot \frac{f_1^2 \cdot f_2^2}{f_1^2 - f_2^2} \cdot \frac{c}{K} \tag{1}$$

where L_1 and L_2 represent the phase observations of GPS signals at f_1 and f_2 frequencies, respectively, c stands for the speed of light in vacuum and $K = 40.3\ \mathrm{m^3/s^2}$.

The ROTI within a 10-minute interval was calculated using the TEC in the dual-frequency GNSS observation data with a time resolution of 30 seconds,

$$ROTI = \sqrt{\langle ROT^2 \rangle - \langle ROT \rangle^2} \tag{2}$$

where,

$$ROT = \frac{sTEC_k^j - sTEC_{k-1}^j}{t_k - t_{k-1}} \tag{3}$$

ROT is the rate of TEC variation in TECU/min,where 10^{16} electrons/m^2= 1 TEC unit (TECU). j denotes the tracked satellite and t_k represents the corresponding epoch time.

Three training models are provided in our software. To help users better understand the mechanisms of the three models, we have compiled the corresponding equations for users to read.

4.1 Xgboost algorithm

XGBoost is an ensemble learning algorithm based on gradient boosting, which gradually builds multiple weak learners (typically decision trees) and combines them to form a strong learner. Each tree attempts to correct the errors of the previous tree, thereby gradually improving the model's accuracy. The mathematical expression of the objective function of XGBoost is as follows:

$$Obj(\theta) = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \Omega(f) \tag{4}$$

where $l(y_i, \hat{y}_i)$ is a loss function that measures the difference between predicted and true values, and its function expression is usually:

$$l(y_i, \hat{y}_i) = -\left[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right] \tag{5}$$

where $\Omega(f)$ is a regularization term defined as:

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \|W\|^2 \tag{6}$$

where T is the number of leaves in the tree, W is the weight vector of leaves, and γ and λ are regularization parameters

XGBoost incrementally incorporates trees into the model during training. Each subsequent tree is trained to minimize the residuals of the preceding trees. The final prediction for a new input$x_i$is the aggregate of the predictions from all trees, as articulated in Eq. (4). The final output is typically passed through a sigmoid function to obtain a probability.

$$\hat{p}_i = \sigma(\hat{y}_i) = \frac{1}{1+exp(-\hat{y}_i)} \tag{7}$$

For binary classification, the predicted class label $\hat{y}_i^{class}$ is:

$$\hat{y}_i^{class} = \begin{cases} 0, if\ \hat{p}_i < 0.5 \\ 1, if\ \hat{p}_i \geq 0.5 \end{cases} \tag{8}$$

4.2 Support vector machines algorithm

For the samples x_i and labels $y_i \in \{-1, +1\}$ in the input space, the classification hyperplane is defined as:

$$w^T \cdot x_i + b = 0 \tag{9}$$

where w is the weight vector and b is the bias term. The geometric interval from the sample point to the hyperplane is:

$$\gamma_i = \frac{y_i(w^T \cdot x_i + b)}{\|w\|} \tag{10}$$

The problem of maximizing the geometric spacing of a plane can be transformed into a problem of minimizing $\|w\|$, which is equivalent to minimizing $\|w\|^2$ under the constraints of Formula (11).

$$y_i(w^T x_i + b) \geq 1, \forall_i \tag{11}$$

Introduce the Lagrange multiplier $\alpha_i \geq 0$ to construct the Lagrange function:

$$\mathcal{L}(\mathrm{w}, \mathrm{b}, \alpha) = \frac{1}{2}\|\mathrm{w}\|^2 - \sum_{i=1}^{n} \alpha_i[y_i(\mathrm{w}^T x_i + b) - 1] \tag{12}$$

By taking the derivative and setting the partial derivative to 0, we obtain:

$$w = \sum_{i=1}^{n} \alpha_i y_i x_i, \sum_{i=1}^{n} \alpha_i y_i = 0 \tag{13}$$

where $\alpha_i \geq 0, \sum_{i=1}^{n} \alpha_i y_i = 0$,Substituting (13) into (12) yields the dual problem:

$$L(\alpha_1 \cdots \alpha_N) = \sum_{i=1}^{n} \alpha_i - \frac{1}{2}\sum_{i,j=1}^{n} \alpha_i \alpha_j y_i y_j x_i \cdot x_j \tag{14}$$

The data is mapped to the high-dimensional space through the nonlinear mapping $\phi(x)$, and the inner product is replaced by the kernel function $K(x_i, x_j)$. At this point, the duality problem becomes:

$$L(\alpha_1 \cdots \alpha_N) = \sum_{i=1}^{n} \alpha_i - \frac{1}{2}\sum_{i,j=1}^{n} \alpha_i \alpha_j y_i y_j K(x_i, x_j) \tag{15}$$

The kernel function used in the RF model of this paper is the Gaussian kernel function. Previous studies have proved that the Gaussian kernel function shows better performance in ionospheric scintillation detection. The Gaussian kernel function is defined as follows:

$$K(x_i, x_j) = exp\left(-\gamma\|x_i - x_j\|^2\right) \tag{16}$$

The final classification decision function is:

$$f(x) = sign\left(\sum_{i=1}^{n} \alpha_i y_i K(x_i, x) + b\right) \tag{17}$$

4.3 Random Forest algorithm

Random forest is an ensemble learning algorithm that improves the accuracy and robustness of the model by constructing multiple decision trees and combining their prediction results. Specifically, it utilizes the bagging algorithm to reduce variance and randomly selects features when the nodes of each decision tree split, thereby increasing the diversity of the model and avoiding overfitting. For the construction of each decision tree, the Gini index or information gain is usually used to select the optimal splitting feature. The Gini index reflects the impurity of the data, and its calculation formula is:

$$Gini(D) = 1 - \sum_{j=1}^{n} p_j^2 \tag{18}$$

D represents the data set, and p_j represents the proportion of the JTH type sample in the data set D. When nodes are split, the feature that satisfies the requirement of minimizing the Gini index is taken as the split feature. For A possible value of feature A, if the data set D is divided into two parts, D1 and D2, then the Gini index under the condition of feature A is:

$$Gini(D, A) = \frac{|D_1|}{D} Gini(D_1) + \frac{|D_2|}{D} Gini(D_2) \tag{19}$$

Information gain represents the improvement in purity after splitting the dataset S using feature F, and its calculation formula is:

$$Gain(F) = Ent(D) - \sum_{v \in F} \frac{|S_v|}{|S|} Ent(S_v) \tag{20}$$

where $Ent(D)$ represents the entropy of the data set D, and the calculation formula is:

$$Ent(D) = -\sum_{j=1}^{n} p_j \, log_2 \, p_j \tag{21}$$

where $Ent(S_v)$ represents the conditional entropy under a certain value v of feature F, and Sv is a subset of the value v of feature F in the dataset S. The greater the information entropy gain is, the stronger the partitioning ability of feature F for the data set S is.

For the classification problem, the prediction results of all decision trees are voted on, and the category with the most votes is taken as the final prediction category. Suppose there is a decision tree in the random forest, and their prediction results are $h_1(\mathrm{x}), h_2(\mathrm{x}), \ldots, h_\mathrm{m}(\mathrm{x})$ respectively, then the final prediction result is:

$$\hat{\mathrm{y}}_\mathrm{i}^{\mathrm{class}} = \underset{\mathrm{y}\in\mathcal{Y}}{\arg\max} \sum_{\mathrm{t}=1}^{\mathrm{m}} \mathrm{I}(h_\mathrm{t}(\mathrm{x}) = \mathrm{y}) \tag{22}$$

where I is the indicator function, and its value is 1 when $h_t(x) = y$, otherwise, it is 0.